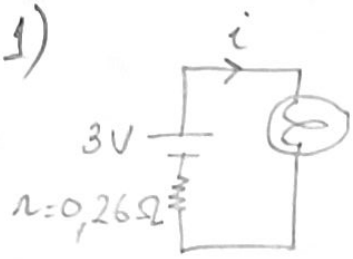
Física 2

GABARITO P2 2019-2

1)

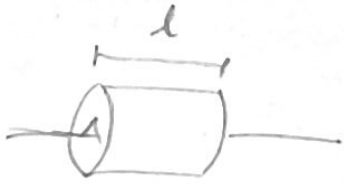


$$\frac{P_R}{P} = \frac{i^2 R}{i^2(r+R)} = \frac{5}{5,26} \sim 95\%$$

2) $R = \rho \frac{l}{A} = \frac{1}{\sigma}\frac{l}{A}$

- Como $R_1 > R_2$

$\Rightarrow \sigma_1 < \sigma_2$



- Como os resistores estão em série: $i_1 = i_2 = i \Rightarrow j_1 = \frac{i_1}{A} = \frac{i_2}{A} = j_2$

- Como $j = \sigma E \Rightarrow \frac{E_1}{E_2} = \frac{j_1/\sigma_1}{j_2/\sigma_2} = \sigma_2/\sigma_1 > 1 \Rightarrow E_1 > E_2$

3)

- Tendo em vista fios ideais: $V_{\pm} = V_C = 0$
- Ao passar por uma bateria no sentido $V_- = V_C = V_{\pm} \to V_A \Rightarrow \Delta V_\varepsilon = \varepsilon$

Portanto, $V_A - \cancel{V_{\pm}}^{0} = \varepsilon \Rightarrow V_A = \varepsilon = 12V$

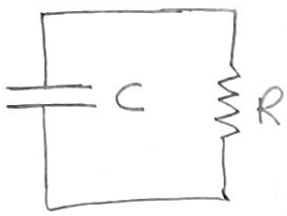
- $\Delta V_{BC} = V_C - V_B = -i R_{2//6}$

onde $R_{eq} = R_{4,5} + R_{2//6} = 4,5 + \frac{2+6}{2\cdot 6} = 4,5 + 1,5 = 6\Omega$

e $\Delta V_\varepsilon + \Delta V_{Req} = 0 \Rightarrow \varepsilon - i R_{eq} = 0 \Rightarrow i = \frac{\varepsilon}{R_{eq}} = \frac{12}{6} = 2A.$

Portanto, $V_B = i R_{2//6} + \cancel{V_{\varepsilon}}^{0} = 2 \cdot 1,5 = 3V.$

(4)



Descarga do capacitor

$$Q(t) = Q_0 e^{-t/RC}$$

$$\Downarrow$$

$$i(t) = \frac{d}{dt} Q(t) = -\underbrace{\frac{Q_0}{RC}}_{i_{máx}} e^{-t/RC}$$

$$i_{MÁX} = \frac{Q_0}{RC}$$

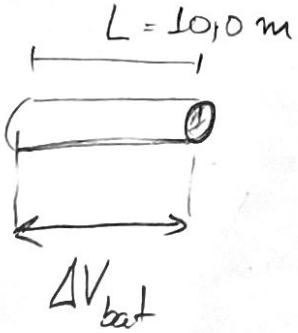
Como $U = \frac{Q^2}{2C} \Rightarrow$ $i_{MÁX} = \frac{\sqrt{2CU}}{RC}$

$$= \frac{\sqrt{2 \cdot 2 \cdot 10^{-6} \cdot 10^{-4}}}{20 \cdot 2 \cdot 10^{-6}}$$

$$= \frac{1}{20} \cdot (10^{-10})^{1/2} \cdot 10^{+6}$$

$$= \frac{1}{20} \cdot 10^{-5+6} = \frac{10}{20} = 0{,}5\,A$$

(5)



$j = 2{,}5 \times 10^6\ A/m^2$

$j = \sigma E$

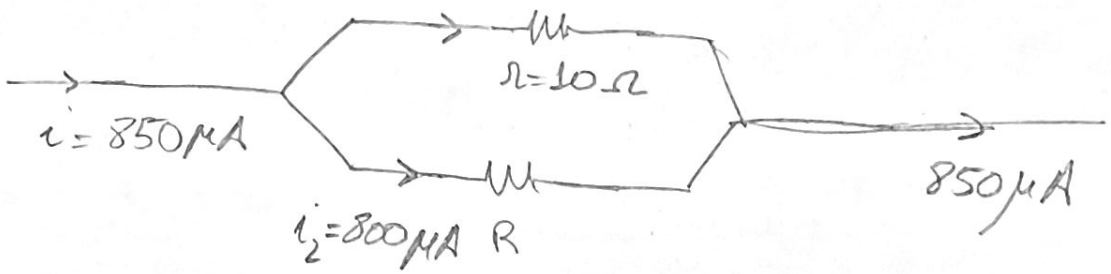
Como $\Delta V = Ed$, $j = \frac{1}{\rho} E = \frac{1}{\rho} \frac{\Delta V}{d} \Rightarrow \rho = \frac{\Delta V}{j d}$

Portanto, $\rho = \frac{1{,}0\,V}{2{,}5 \times 10^6 \cdot 10} = \frac{1}{\frac{5}{2} \times 10^7} = \frac{2}{5} \times 10^{-7} = \frac{20}{5} \times 10^{-8}$

$= 4{,}0 \times 10^{-8}\ \Omega m$

$\Delta V = i_1 \cdot \Omega = 50 \cdot 10 = 500\,\mu V$ (3)

(6)



Portanto, no caso limite

$$\Delta V = i_2 \cdot R$$

$$500 \times 10^{-6} = 800 \times 10^{-6} R$$

$$R = \frac{5}{8} = 0{,}625\,\Omega.$$

50 | 8
10, 625
20
40

De forma a manter ΔV = cte, se R aumenta, i_2 diminui. Todavia, como $i = i_1 + i_2$, se $i_2 \downarrow \Rightarrow i_1 \uparrow$.

(7) $P = i^2 R$

$50 = 1^2 \cdot R$

$R_L = 50\,\Omega$

$\Delta V_R = -iR$

$= -1 \cdot 20 = -20V = V_C - V_A$

$\Delta V_L = -i\,50 = -50V = V_B - V_C$

Portanto, $\Delta V_R + \Delta V_L = (V_C - V_A) + (V_B - V_C) = V_B - V_A = \Delta V_{AB}$

$-20 - 50 = \Delta V_{AB}$

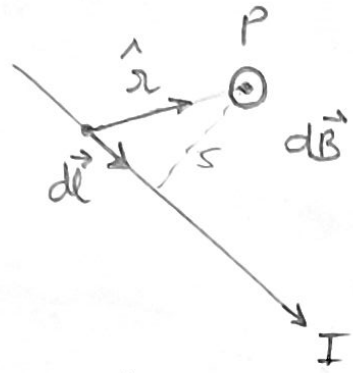
$\Delta V_{AB} = -70V$

8- fios retilíneos muito longo = fio ∞

$$B_{fio\infty} = \frac{\mu_0 i}{2\pi s}$$

Pelo princípio da superposição: $B_p = 2\,B_{fio\infty} = \frac{2\mu_0 i}{2\pi s} = \frac{\mu_0 i}{\pi s}$

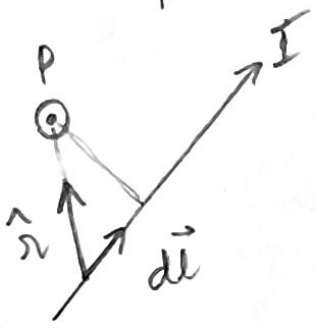
fio 1



$d\vec{B} = \frac{\mu_0 i}{4\pi} \frac{d\vec{l}\times\hat{r}}{s^2}$ p/ fora da folha

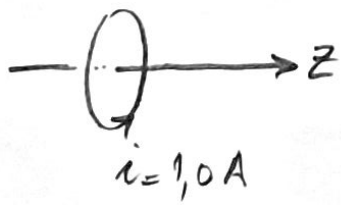
+

fio 2



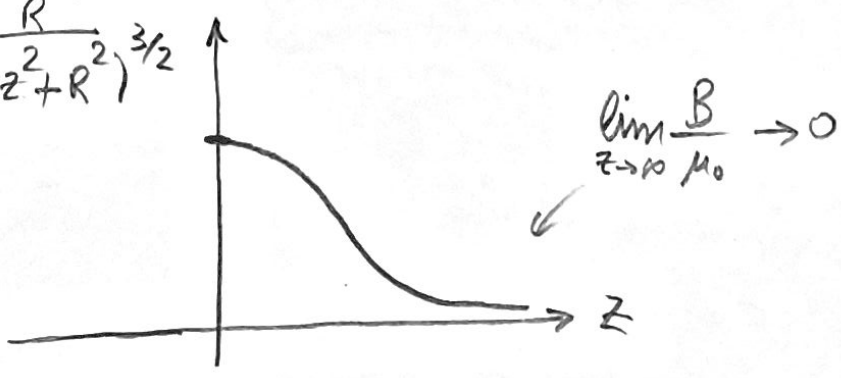
p/ fora da folha

9-



$\therefore B_{espira} = \frac{\mu_0 i R^2}{2(z^2+R^2)^{3/2}}$

Portanto, $\frac{B}{\mu_0} = \frac{R^2}{2(z^2+R^2)^{3/2}}$



$\frac{B}{\mu_0}(z=0) = \frac{(10^{-2})^2}{2(10^{-2})^{3/2}}$

$= \frac{1}{2}\frac{10^{-4}}{10^{-3}}$

$= 0{,}5\times10^{-1}$

13- $B_{bob} = N B_{arco\ 2\pi}$

$$= \frac{\mu_0 i N \cancel{2\pi}}{2 \cancel{4\pi} d}$$

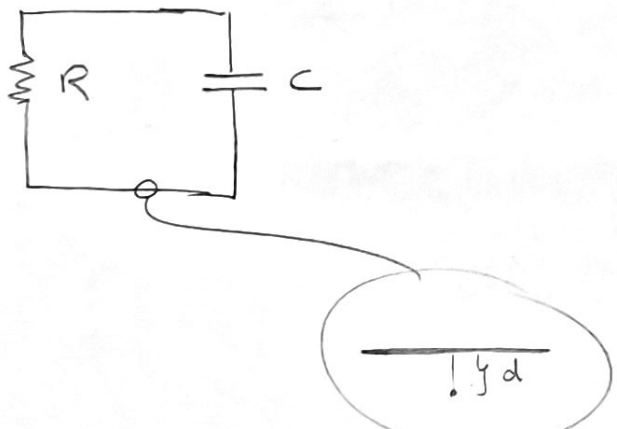
$$2{,}0\pi \times \cancel{10^{-6}} = \frac{\mu_0 \cdot 100 \times \cancel{10^{-6}}\, N}{2\ 10^{-2}}$$

$$N = \frac{4\pi \times 10^{-4}}{\mu_0} = \frac{\cancel{4\pi}}{\cancel{4\pi} \times 10^{-7}} \cdot 10^{-4} = 10^3 = 1000 \text{ voltas}$$

$$\Downarrow$$

1000 espiras !!

14-



$\vec{B}$ Muito proximo do fio $= \vec{B}_{fio\ \infty}$: $B_{fio\ \infty} = \frac{\mu_0 i}{2\pi d}$

contudo; $i = \frac{d}{dt} Q(t) = -\underbrace{\frac{Q_0}{RC}}_{i_0} e^{-t/RC} = -i_0 e^{-t/RC}$

$$i(T) = \frac{\cancel{i_0}}{2} = \cancel{i_0}\, e^{-T/RC}$$

$$\ln \tfrac{1}{2} = -T/RC$$

$$\cancel{\ln(1)} + \ln(2) = +T/RC \quad (\ln(1) = 0)$$

$$T = RC \ln(2)$$

$$= 2 \times \cancel{10^{3}} \times 5 \times \cancel{10^{-3}} \ln(2)$$

$$= 10 \ln(2)$$

$$= 10 \cdot 0{,}693$$

$$= 6{,}93 \sim 7\,\Omega$$